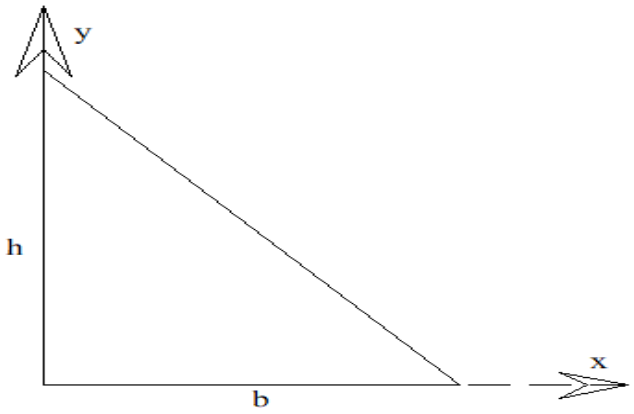
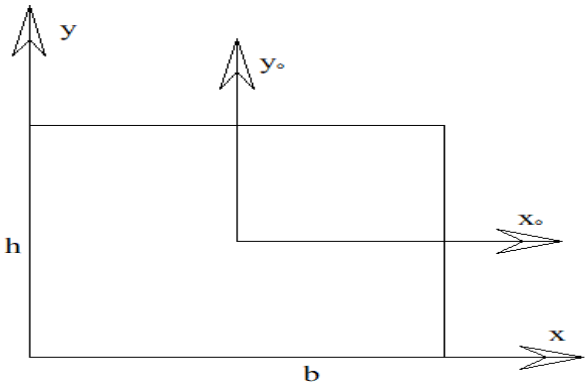
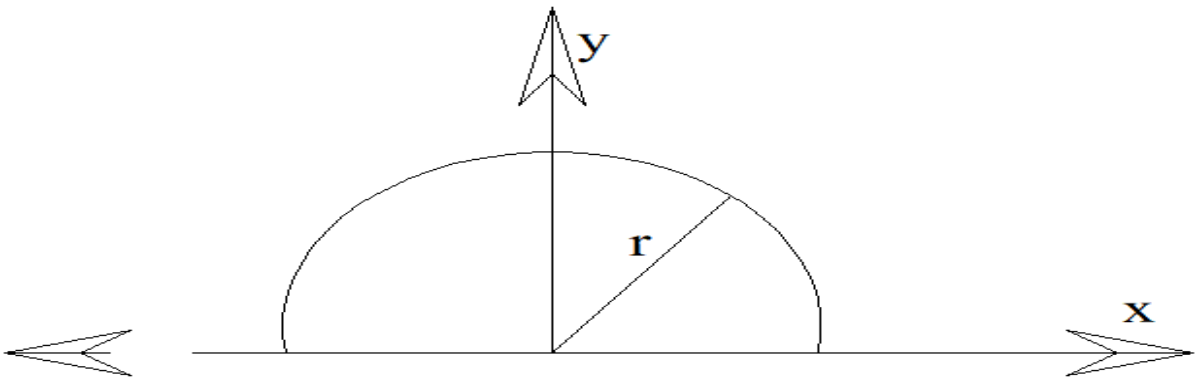
1. Determine the coordinates of centroid of right angle triangle and its moment of inertia about the centroidal axis and x- y axis.



2. Determine the centroid and moment of inertia of the rectangular area.
   a. About x – y axis
   b. About centroidal (x_o – y_o) axis



3. Determine the coordinates of the centroid of the half circle area and its moment of inertia about x-y axis and centroidal axis.



4. Determine the coordinates of centroid of the quarter circle (Fig.1a) area and its moment of inertia about x-y axis and centroidal axis.
5. Determine the centroid of the sector area (Fig.1b).

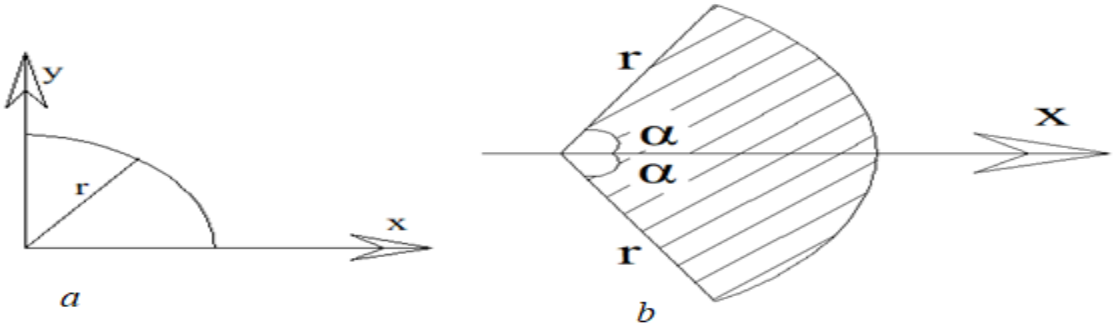


Figure 1: Quarter and sector area

6. If the diagonal of triangle and line OB which pass through center of quarter circle are collinear as shown in Fig.2, determine the coordinates centroid of the composite body measured from $x_o - y_o$.
   B. Moment of inertia about centroidal axis and x – y axis shown on Fig.2 using transfer axis theorem.

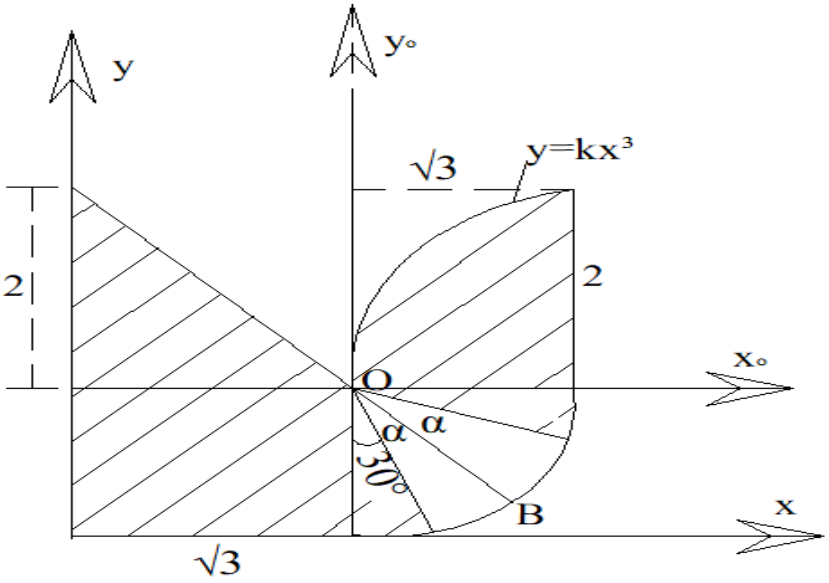


Figure 2: Composite body for Q #6

7. Determine the x-y coordinates centroid of the composite body shown in Fig.3. and its moment of inertia about centroidal axis and x – y axis using transfer axis theorem.

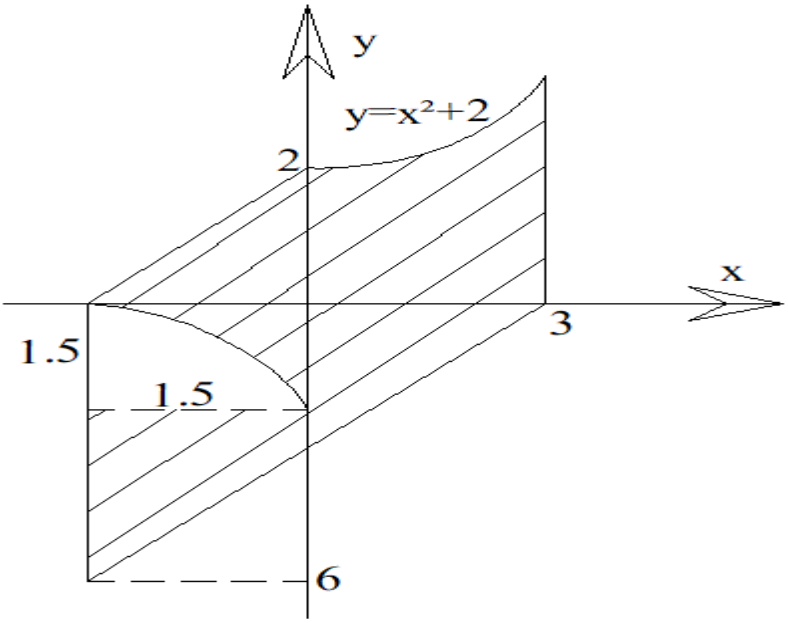


Figure 3: Composite body

8. For the beam shown in Fig.4a & b
   a. Draw the shear force diagram (SFD) and bending moment diagram (BMD) and indicate the values at inflection points.
   b. Determine the maximum moment and maximum shear force and the corresponding point the segment of beam.
   c. Calculate the bending moment at point where shear force is zero.
   d. Show the relationship of the load, shear and bending moment using the results of internal force analysis.

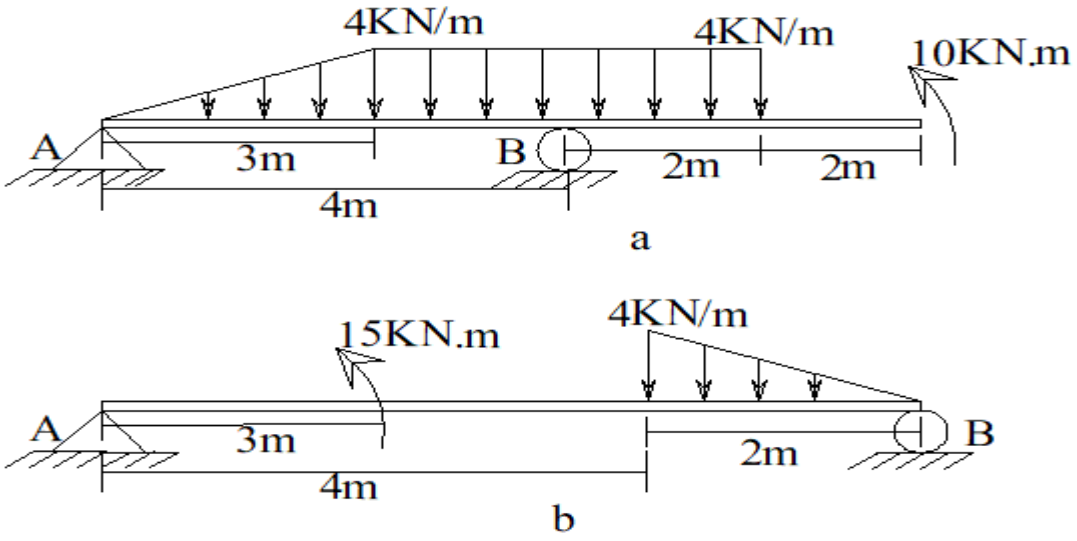


Figure 4: Beam subjected to distributed load